3.3.1, there are exactly r irreducible representations of A (isomorphic to $k^{d_1}, \dots, k^{d_r}$, respectively) and therefore r linearly independent characters on the r-dimensional vector space $A/[A, A]$. Thus, the characters form a basis. $\square$

3.7. The Jordan-Hölder theorem

We will now state and prove two important theorems about representations of finite dimensional algebras — the Jordan-Hölder theorem and the Krull-Schmidt theorem.

Theorem 3.7.1 (Jordan-Hölder theorem)**.** *Let V be a finite dimensional representation of A, and let $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$, $0 = V'_0 \subset \cdots \subset V'_m = V$ be filtrations of V, such that the representations $W_i := V_i/V_{i-1}$ and $W'_i := V'_i/V'_{i-1}$ are irreducible for all i. Then $n = m$, and there exists a permutation σ of $1, \dots, n$ such that $W_{\sigma(i)}$ is isomorphic to W'_i.*

Proof. *First proof* (for k of characteristic zero). The character of V obviously equals the sum of characters of W_i and also the sum of characters of W'_i. But by Theorem 3.6.2, the characters of irreducible representations are linearly independent, so the multiplicity of every irreducible representation W of A among W_i and among W'_i is the same. This implies the theorem.[5]

Second proof (general). The proof is by induction on $\dim V$. The base of induction is clear, so let us prove the induction step. If $W_1 = W'_1$ (as subspaces), we are done, since by the induction assumption the theorem holds for V/W_1. So assume $W_1 \neq W'_1$. In this case $W_1 \cap W'_1 = 0$ (as W_1, W'_1 are irreducible), so we have an embedding $f : W_1 \oplus W'_1 \to V$. Let $U = V/(W_1 \oplus W'_1)$, and let $0 = U_0 \subset U_1 \subset \cdots \subset U_p = U$ be a filtration of U with simple quotients $Z_i = U_i/U_{i-1}$ (it exists by Lemma 3.4.2). Then we see the following:

1) V/W_1 has a filtration with successive quotients $W'_1, Z_1, \dots, Z_p$ and another filtration with successive quotients $W_2, \dots, W_n$.

[5]This proof does not work in characteristic p because it only implies that the multiplicities of W_i and W'_i are the same modulo p, which is not sufficient. In fact, the character of the representation pV, where V is any representation, is zero.